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GROUP-BASED FILE SHARING APPLICATION

Network Programming - IT4062E - 1st Progress Report

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ONE LOVE. ONE FUTURE.

1. Actors & Problem Definition

System Actors

- **Regular Member**:
  - Registers & signs in to the system.
  - Requests to join groups.
  - Uploads/downloads files, creates subfolders in group space.
- **Group Admin** (Group Creator):
  - Creates and manages sharing groups.
  - Exclusive rights: rename & delete files and subfolders.

Problem Definition

- **Group-Centric Collaboration**: Provide dedicated shared folders for teams to organize and collaborate online.
- **Hierarchical Content Organization**: Allow members to structure project data with nested subfolders and file uploads.
- **Role-Based Access Control**: Enforce administrative authority so that only group admins can rename or delete files and folders, preventing unauthorized data loss.
- **Network File Transfer**: Support secure client-server file uploads and downloads across the network.

2. Core Use Cases

Account & Group Management

- **UC1 — Register & Log In**
  Register a new account and authenticate for usage.
- **UC2 — Create Sharing Group**
  Create a new group with a dedicated shared folder; creator gains admin privileges.
- **UC3 — Join Group**
  Browse existing groups and submit requests to join.

File & Folder Operations

- **UC4 — Upload & Download Files**
  Any group member can upload files to the group directory or download files to local storage.
- **UC5 — Create Subfolders**
  Group members can create subfolders inside the group's shared folder.
- **UC6 — Manage Files & Folders (Admin)**
  Exclusive authority to delete files/subfolders and rename files/folders.

3. Inputs & Outputs Specification

Account & Group Operations

- **Account Registration & Login**:
  - *In:* Username, password credentials.
  - *Out:* Auth status (success/failure), session token.
- **Create Sharing Group**:
  - *In:* Group name, creator user ID.
  - *Out:* Group created, dedicated folder provisioned, admin privileges assigned.
- **Request to Join Group**:
  - *In:* Target Group ID, requesting user ID.
  - *Out:* Join request status (pending/approved), member list updated.

File & Directory Operations

- **File Upload & Download**:
  - *In:* Group/folder path, file stream/metadata (upload); file path (download).
  - *Out:* File saved / streamed to client; transfer status.
- **Create Subfolder**:
  - *In:* Target parent path, new subfolder name.
  - *Out:* Subfolder created; directory view updated.
- **Rename & Delete (Admin Only)**:
  - *In:* Admin session, target item path, new name (rename) or deletion signal.
  - *Out:* Permission validated; item renamed or deleted; updated tree returned.



THANK YOU !